

Homework 3

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Deadline: 2022-12-16 22:30:00, Beijing time

How to submit?

Email: YJ40202@163.com

Subject: [homework](#)

Attachments:

DecimalToBinary.java, Bubble.java, Merge.java and Matrix.java

Note: Don't use package and don't submit compressed file.

How to submit?

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DecimalToBinary.java

Write a method that converts a decimal number into a binary number as a string. The method header is:

```
public static String dec2Bin(int value)
```

Write a test program that prompts the user to enter a decimal number and displays its binary equivalent.

Don't use Java's `Integer.toString(int)` in this program.

Sample runs

```
Enter an integer: 12345
12345's binary representation is 11000000111001
```

Bubble.java

Write down a sort method to sort an array of integers using bubble-sort algorithm. Bubble sort algorithm makes several passes through the array. On each pass, successive neighboring pairs are compared. If a pair is not in order, its values are swapped; otherwise the value remains unchanged.

```
static void bubbleSort(double[] list)
```

Write a test program that reads in 10 integers and invokes the bubble sort method. Display the sorted int array.

Sample run

```
Enter 10 numbers: 1 9 2 5 8 10 2 4 3 3  
The sorted array is: 1.0 2.0 2.0 3.0 3.0 4.0 5.0 8.0 9.0 10.0
```

Merge.java

Write the following method that merges two sorted lists into a new sorted list:

```
public static int[] merge(int[] list1,int[] list2)
```

Write a test program that prompts the user to enter two sorted lists and displays the merged list.

Here is a sample run.

```
Enter list1 size and contents: 5 1 5 16 61 111
Enter list2 size and contents: 4 2 4 5 6
The merged list is:1 2 4 5 5 6 16 61 111
```

Matrix.java

Write a method to multiply two matrices. The header of the method is:

```
public static double[][] multiplyMatrix(double [][] a, double [][] b).
```

To multiply matrix a by matrix b, the number of columns in a must be the same as the number of rows in b, and the two matrices must have elements of the same or compatible types. Let c be the result of the multiplication. Assume the column size of matrix a is n. Each element $c_{ij} = a_{i1} \times b_{1j} + a_{i2} \times b_{2j} + \dots + a_{in} \times b_{nj}$. For example, the two 3×3 matrices a and b, c is:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix}$$

where $c_{ij} = a_{i1} \times b_{1j} + a_{i2} \times b_{2j} + a_{i3} \times b_{3j}$.

Matrix.java

Write a test program that prompts the user to enter two 3×3 matrices and displays their product.

Here is a sample run:

```
Enter matrix1: 1 2 3 4 5 6 7 8 9
Enter matrix2: 0 2 4 1 4.5 2.2 1.1 4.3 5.2
The multiplication of the matrices is
  5.300000    23.900000    24.000000
 11.600000    56.300000    58.200000
 17.900000    88.700000    92.400000
```